

AC2206 & AC3302

Investment Appraisal

Winter 2024 Examination

Paper 1 · 1.5 hours · Answer Question 1 and one other question · Total 100 Marks

Module	AC2206 & AC3302 — Investment Appraisal
Session	Winter 2024
Paper	1 of 1
External Examiner	Professor Thomas O'Connor
Internal Examiner	Mr. David Humphreys
Structure	Question 1 (Compulsory, 50 Marks) + one of Question 2 or Question 3 (50 Marks)
This scheme covers	Question 1, Question 2 and Question 3 in full, so you can study either route through the paper

Own-figure rule. Where a candidate makes an error early in a multi-step question, full method marks are still awarded for correctly applying the required technique to their own, incorrect figure in every subsequent step. This applies throughout every question in this scheme.

Quality check. Every calculation in this scheme has been independently verified and cross-checked for accuracy before being written up.

Question 1 (Compulsory)

50 Marks — NPV with Tax, Capital Allowances & Working Capital

Scenario

Oakmont plc is considering a €20 million investment in plant and machinery. Sales: €35m in year 1, dropping to €20m in year 2, then rising 10% p.a. thereafter. The plant is scrapped after 5 years with a salvage value of €1 million. Capital allowances: 40% in year 1, then 20% reducing balance. Operating costs are 75% of sales. A €1,000,000 increase in working capital is required. Tax at 30%, paid one year in arrears. Cost of capital 10%. All cash flows occur at year end.

(a) Required

30 Marks

Using the NPV method of project appraisal, recommend whether the project should be accepted.

FULL MODEL ANSWER

Sales, operating costs and contribution

Year	Sales	Operating Costs (75%)	Contribution (25%)
1	€35.000m	€26.250m	€8.750m
2	€20.000m	€15.000m	€5.000m
3 (×1.10)	€22.000m	€16.500m	€5.500m
4 (×1.10)	€24.200m	€18.150m	€6.050m
5 (×1.10)	€26.620m	€19.965m	€6.655m

Reading the sales pattern. Sales fall from €35m in year 1 to €20m in year 2 (a one-off drop, likely reflecting an initial launch spike followed by normalisation), and only **from year 3 onward** do sales grow at 10% p.a. on the new, lower year-2 base — not from year 1. This scheme applies the 10% growth strictly to years 3–5 as stated ("thereafter" refers to after the year-2 drop).

Question 1 (continued)

Capital allowances (40% year 1, then 20% reducing balance, balancing allowance in year 5)

Year	Opening TWDV	Allowance	Closing TWDV
1	€20.000m	$40\% \times 20.000 = \text{€}8.000\text{m}$	€12.000m
2	€12.000m	$20\% \times 12.000 = \text{€}2.400\text{m}$	€9.600m
3	€9.600m	$20\% \times 9.600 = \text{€}1.920\text{m}$	€7.680m
4	€7.680m	$20\% \times 7.680 = \text{€}1.536\text{m}$	€6.144m
5 (disposal, €1m salvage)	€6.144m	Balancing allowance = $6.144 - 1.000 = \text{€}5.144\text{m}$	€0.000m
Total capital allowances		€19.000m	

Total capital allowances (€19.000m) exactly equal the cost of the asset less its salvage value (€20m – €1m = €19m) — the expected reconciliation once the year-5 balancing allowance is included, confirming no value has been lost or double-counted.

Taxable profit and tax (30%, one year in arrears)

Year	Contribution	Less: Capital Allowance	Taxable Profit	Tax @ 30%	Tax Paid in Year
1	€8.750m	€8.000m	€0.750m	€0.225m	2
2	€5.000m	€2.400m	€2.600m	€0.780m	3
3	€5.500m	€1.920m	€3.580m	€1.074m	4
4	€6.050m	€1.536m	€4.514m	€1.354m	5
5	€6.655m	€5.144m	€1.511m	€0.453m	6

Question 1 (continued)

FULL MODEL ANSWER

The €1,000,000 increase in working capital is invested at $t = 0$ and, since the plant is scrapped (production ends) at the end of year 5, is assumed to be released and recovered as a cash inflow at $t = 5$, alongside the €1m salvage value.

Year (t)	Contribution CF	Other Cash Flows	Tax Paid	Net Cash Flow	PV Factor @ 10%	Present Value
0	—	Plant (–€20.000m) + WC (–€1.000m)	—	— €21.000m	1.000	— €21.000m
1	€8.750m	—	€0.000m	€8.750m	0.909	€7.954m
2	€5.000m	—	€0.225m	€4.775m	0.826	€3.944m
3	€5.500m	—	€0.780m	€4.720m	0.751	€3.545m
4	€6.050m	—	€1.074m	€4.976m	0.683	€3.399m
5	€6.655m	Salvage (+ €1.000m) + WC recovery (+ €1.000m)	€1.354m	€7.301m	0.621	€4.534m
6	—	—	€0.453m	— €0.453m	0.564	— €0.256m
NPV						+€2.12m

Recommendation. The project has a modest but positive NPV of approximately **+€2.12 million**. **Oakmont plc should accept the project**, since it is expected to create shareholder value at the required 10% return — though the margin is much tighter than a typical NPV problem in this series, so the recommendation is more sensitive to the underlying assumptions (see note below) than usual.

Assumptions flagged. (1) The question does not explicitly state whether the working capital is recovered — this scheme assumes it is recovered in full at $t = 5$, the standard convention when a project has a defined end date and no stated write-off; a candidate who instead treated working capital as unrecovered would correctly get a lower NPV (own-figure). (2) The closing sentence noting that Oakmont "expects...it will be replaced by a similar investment" is scene-setting context about the company's longer-term capital planning; it does not change this single-project NPV calculation (a replacement-chain or equivalent-annual-cost analysis is not required, since the question asks only for the NPV of accepting or rejecting this project).

Mark Allocation — Part (a), 30 Marks

Tier	Marks	What separates this tier
24-30	24-30	Correct sales pattern (drop then 10% growth from year 3), correct capital allowance schedule including year-5 balancing allowance, correct one-year tax lag, correct treatment of salvage value and working capital recovery in year 5, and correct final NPV with a clear recommendation.
16-23	16-23	Correct overall structure with one component error — e.g. 10% growth applied from year 1 instead of year 3, or working capital assumed not recoverable (own-figure carried through, with the assumption clearly stated).
8-15	8-15	Contribution and capital allowances calculated but tax timing, salvage value, or working capital not correctly incorporated.
0-7	0-7	No credible attempt to incorporate tax, capital allowances or working capital into the cash flow.

Question 1 (Compulsory)

50 Marks — NPV with Tax, Capital Allowances & Working Capital

(b) Required

20 Marks

"Despite theoretical inferiorities industry practitioners are more likely to use simplistic methods of investment appraisal rather than those methods whose assumptions are considered more academically rigorous." Discuss the above statement paying particular attention to the findings of any empirical research you consider relevant.

FULL MODEL ANSWER

Survey research into corporate investment appraisal practice (e.g. Pike 1996 in the UK; Graham & Harvey 2001 in the US; Arnold & Hatzopoulos 2000 in the UK) provides considerable, though not unqualified, support for this statement.

Evidence supporting the statement — persistent use of payback period:

Despite ignoring the time value of money (in its simple form) and ignoring cash flows beyond the payback point, payback period remains one of the most widely used appraisal techniques in practice, frequently used even by firms that also compute NPV or IRR — a clear real-world preference for a simpler, more intuitive method over the theoretically superior discounted cash flow techniques.

Why simplistic methods persist despite theoretical inferiority: Several genuine, defensible reasons help explain this rather than it simply reflecting poor practice: payback is easy to compute and communicate to non-financial managers and board members; it provides a rough, intuitive proxy for liquidity risk and how quickly capital is recovered, which matters independently of pure value maximisation, especially for firms facing capital rationing; and given that long-term cash flow forecasts are inherently uncertain, the apparent extra 'precision' of a fully discounted NPV model can be somewhat illusory ('garbage in, garbage out'), making a simpler heuristic a rational response to genuine forecast uncertainty rather than a purely irrational one.

Evidence qualifying/challenging the statement — DCF usage has risen

substantially: More recent survey evidence shows a marked increase over time in the proportion of large firms using NPV and/or IRR as a primary appraisal method, particularly among larger, more financially sophisticated organisations — suggesting the gap between theory and practice has narrowed considerably rather than being a fixed or worsening feature of practice.

Firms typically use multiple techniques together: Rather than choosing simplistic methods instead of rigorous ones, most firms in practice apply several methods simultaneously (e.g. NPV alongside payback and accounting rate of return), using each to answer a different question (value creation, liquidity/risk, and reported accounting performance respectively) — this nuances the statement, since it is not simply a case of 'simplistic instead of rigorous' but often 'simplistic alongside rigorous'.

Balanced conclusion: The statement holds true in the specific sense that payback period and other simple heuristics remain stubbornly popular despite well-documented theoretical weaknesses, but it somewhat overstates the case if read as implying rigorous DCF methods are being abandoned or under-used — the empirical picture is one of increasing DCF sophistication in large firms, used in combination with, rather than strictly instead of, simpler measures.

Candidates who engage critically with the statement (rather than either simply agreeing or simply disagreeing) — citing specific empirical findings on payback persistence and/or rising DCF adoption, and offering a reasoned explanation for why simpler methods persist — should receive full credit regardless of their final position.

Mark Allocation — Part (b), 20 Marks

Tier	Marks	What separates this tier
16-20	16-20	Critical, balanced engagement with the statement, citing specific empirical findings (payback persistence AND rising DCF adoption), with a well-reasoned explanation for why simpler methods persist despite theoretical inferiority.
11-15	11-15	Good discussion of empirical findings but one-sided (only supports or only challenges the statement) or explanation for persistence underdeveloped.
6-10	6-10	General discussion of appraisal techniques with limited or no reference to empirical research.
0-5	0-5	No meaningful engagement with the statement or the empirical literature.

Question 1 Total: 50 Marks

Question 2

50 Marks — Choice question: answer EITHER Question 2 OR Question 3

How to use this scheme. The paper requires Question 1 plus **one** of Question 2 or Question 3 — both are marked out of 50 and count equally. This scheme covers both in full so you can revise whichever route you are more comfortable with.

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Question 2

50 Marks — Choice question: answer EITHER Question 2 OR Question 3

Scenario

Southern Hills plc has €750,000 available to invest across six possible, non-divisible projects that cannot be postponed. Required rate of return: 12%.

Project	Year 1	Year 2	Year 3	Year 4	Year 5	Initial Outlay
A	80	80	80	80	80	279
B	75	85	64	—	—	155
C	48	48	63	73	—	175
D	62	62	62	62	—	195
E	40	50	60	70	40	170
F	35	82	82	—	—	160

All cash flows in €000s.

(a) Required

10 Marks

Calculate the expected net present value of each project.

FULL MODEL ANSWER

Project	PV of Cash Inflows (12%)	Initial Outlay	NPV
A	288.40	279	+9.40
B	180.29	155	+25.29
C	172.40	175	−2.60
D	188.36	195	−6.64
E	185.49	170	+15.49
F	154.99	160	−5.01

All figures in €000s, using the 12% present value factors provided. Each project's PV of inflows was computed independently in Python and cross-checked against the exact (unrounded) discount factors, with no material difference — the table-based figures above are used for consistency with the factors provided in the paper.

Mark Allocation — Part (a), 10 Marks

Tier	Marks	What separates this tier
9-10	9-10	All six NPVs correct, with correct sign (positive/negative) in every case.
6-8	6-8	Four or five NPVs correct.
3-5	3-5	Two or three NPVs correct, or systematic error (e.g. wrong discount factors) applied consistently.
0-2	0-2	Method not recognised or no credible attempt.

Question 2

50 Marks — Choice question: answer EITHER Question 2 OR Question 3

(b) Required

10 Marks

Calculate the expected profitability index of each project.

FULL MODEL ANSWER

Profitability Index (PI) = PV of Cash Inflows ÷ Initial Outlay

Project	PV of Cash Inflows	Initial Outlay	Profitability Index
A	288.40	279	1.034
B	180.29	155	1.163
C	172.40	175	0.985
D	188.36	195	0.966
E	185.49	170	1.091
F	154.99	160	0.969

A profitability index above 1.0 indicates a positive NPV (PV of inflows exceeds outlay), and below 1.0 indicates a negative NPV — consistent with the sign of each project's NPV in part (a). PI expresses value created per euro invested, which is the basis for ranking under capital rationing in part (c).

Mark Allocation — Part (b), 10 Marks

Tier	Marks	What separates this tier
9-10	9-10	All six profitability indices correct (own-figure from part (a) if applicable).
6-8	6-8	Four or five correct.
3-5	3-5	Two or three correct, or PI formula inverted/misapplied consistently.
0-2	0-2	Method not recognised.

Question 2

50 Marks — Choice question: answer EITHER Question 2 OR Question 3

(c) Required

5 Marks

Rank the projects according to both of these investment appraisal methods and make a recommendation as to which projects should be selected.

FULL MODEL ANSWER

Rank	By NPV	By Profitability Index
1	B (+25.29)	B (1.163)
2	E (+15.49)	E (1.091)
3	A (+9.40)	A (1.034)
4	C (−2.60)	C (0.985)
5	F (−5.01)	F (0.969)
6	D (−6.64)	D (0.966)

Ranking is identical under both methods here: Because all six projects are being compared on a common basis (all under the same required rate of return, and PI is simply NPV per euro of outlay rescaled), the NPV ranking and PI ranking happen to produce exactly the same order in this case: B, E, A, C, F, D.

Recommendation

Selected (in PI rank order)	Outlay (€000s)	Cumulative Outlay	NPV (€000s)
B	155	155	+25.29
E	170	325	+15.49
A	279	604	+9.40
Total selected		604 (of 750 available)	+50.18

Recommendation: select Projects B, E and A — the three positive-NPV projects, which together cost €604,000 (within the €750,000 budget, leaving €146,000 unused since Projects C, D, and F all have negative NPV and should not be undertaken regardless of remaining budget). This selection was verified by exhaustively checking every combination of projects that fits within the €750,000 budget: no other affordable combination produces a higher total NPV than €50.18k, confirming that simply ranking by profitability index and selecting down the list gives the value-maximising choice in this case. This will not always hold exactly for indivisible projects — profitability index ranking is only guaranteed optimal when projects are perfectly divisible — but it does hold here.

Mark Allocation — Part (c), 5 Marks

Tier	Marks	What separates this tier
5	5	Correct ranking under both methods, and a recommendation to select B, E and A (or the correct own-figure equivalent) with a clear budget check.
3-4	3-4	Correct ranking but recommendation includes a negative-NPV project or omits a positive-NPV one within budget.
1-2	1-2	Ranking attempted but recommendation not clearly tied to the €750,000 budget constraint.
0	0	No credible ranking or recommendation.

Question 2

50 Marks — Choice question: answer EITHER Question 2 OR Question 3

(d) Required

25 Marks

"Risk analysis has become an increasingly important part of the investment appraisal process". Discuss the above statement paying particular attention to any methods of risk assessment that you have studied. To what extent do you believe that these methods are effective in measuring the risk associated with an investment project?

FULL MODEL ANSWER

Risk analysis has indeed grown in prominence in investment appraisal, reflecting greater recognition that a single-figure NPV based on expected cash flows can mask significant underlying uncertainty. Several methods are commonly studied and applied:

Sensitivity analysis: Tests how NPV changes as one input variable (e.g. sales volume, price, discount rate) is varied in isolation, holding others constant, to identify which variables the project's viability is most sensitive to. It is simple to apply and communicates clearly which assumptions matter most, but its key weakness is that it varies only one input at a time and so ignores realistic correlations between variables (e.g. price and volume often move together).

Scenario analysis: Extends sensitivity analysis by varying several inputs together under coherent 'best case / base case / worst case' scenarios, giving a more realistic picture of how combinations of adverse conditions might affect NPV. It is more realistic than single-variable sensitivity analysis but is still limited to a small number of discrete scenarios and does not give a full probability distribution of outcomes.

Probability/expected value analysis and decision trees: Assigning explicit probabilities to different outcomes (as in Question 1 of this paper's approach to uncertain quantity and demand) allows an expected NPV to be calculated, and decision trees extend this to sequential decisions with multiple stages of uncertainty. This is more rigorous than sensitivity analysis but requires credible probability estimates, which are often themselves highly subjective and difficult to justify with precision.

Standard deviation / coefficient of variation of NPV: Where a full probability distribution of possible NPVs can be estimated, the standard deviation (absolute risk) or coefficient of variation (risk relative to expected return) of NPV gives a formal risk measure that can be compared across projects of different scale — but this requires a well-specified probability distribution, which is difficult to estimate reliably in practice for a novel project.

Monte Carlo simulation: Uses computer simulation to repeatedly sample from probability distributions for each uncertain input, generating a full distribution of possible NPV outcomes rather than a small number of discrete scenarios. It captures interactions between variables more realistically than sensitivity or scenario analysis, but is more complex to build, requires specifying a full distribution (not just a range) for every uncertain input, and its output is only as reliable as those input distributions.

Risk-adjusted discount rate and certainty equivalents: A higher discount rate can be applied to riskier projects (or cash flows converted to their certainty-equivalent, lower-risk value before discounting at the risk-free rate) to build a risk premium directly into the appraisal — simple to apply, but the choice of risk premium is often somewhat arbitrary and can compound the discounting effect on cash flows further in the future in a way that may not reflect how risk actually evolves over the project's life.

To what extent are these methods effective?

Genuine value — but no method eliminates the core problem: All of these methods add real value by moving beyond a single-point NPV estimate and forcing more explicit thinking about uncertainty, which is itself valuable even before any numbers are produced. However, none of them can overcome the fundamental problem that they all still rely on subjective, forward-looking estimates (of ranges, probabilities, or distributions) — more sophisticated risk techniques do not eliminate estimation risk, they simply make its structure and sensitivity more visible.

Trade-off between rigour and practicality: The more rigorous techniques (Monte Carlo simulation, formal probability distributions) provide the richest picture of risk but are the most complex, time-consuming, and dependent on credible input distributions — consistent with the empirical finding (Question 1(b) of this paper) that simpler tools such as sensitivity and scenario analysis are far more widely used in practice than formal simulation, precisely because they offer a reasonable balance of insight and practicality.

Overall assessment: These methods are effective at improving the quality of the decision-making process and communicating risk to decision-makers, but should be seen as aids to judgement rather than a mechanism for objectively 'measuring' true risk with precision — the ultimate reliability of any risk analysis is bounded by the quality of the underlying assumptions fed into it.

Candidates who discuss at least three distinct risk assessment methods with genuine understanding, and offer a reasoned critical assessment of their effectiveness (not simply listing techniques), should receive full credit regardless of which specific methods they emphasise.

Mark Allocation — Part (d), 25 Marks

Tier	Marks	What separates this tier
20-25	20-25	At least four distinct risk assessment methods discussed with genuine understanding of their mechanics and limitations, PLUS a well-reasoned critical assessment of their overall effectiveness.
13-19	13-19	Two to three methods discussed adequately; critical assessment of effectiveness present but underdeveloped.
6-12	6-12	Methods listed/named but with limited explanation of mechanics or limitations; little critical assessment.
0-5	0-5	No meaningful discussion of risk assessment methods.

Question 2 Total: 50 Marks

Question 3

50 Marks — Choice question: alternative to Question 2

(a) Scenario

20 Marks

On October 1 2024 a U.S. investor holds a well-diversified stock portfolio with a market value of \$5 million, constructed to match the performance of the S&P 500 index. Worried about a significant drop in equity prices over the following six-month period, the investor is considering the options market to hedge.

On October 1 2024 the S&P 500 index stands at 4300. Available S&P 500 index options (all with a \$500 multiplier): December 2024 expiration, strike 4250; March 2025 expiration, strike 4300; June 2025 expiration, strike 4350.

- (i) Recommend a suitable hedging strategy for the investor. (10 Marks)
- (ii) Calculate the payoff to such a strategy in the event that the S&P 500 declines by 15% over the next six months. (10 Marks)

FULL MODEL ANSWER

(i) Recommended strategy: Buy S&P 500 index **put options** — a protective put on the portfolio, which fully tracks the S&P 500. This locks in a floor value for the portfolio while preserving upside if the market rises instead.

Choice of expiration: the investor's stated horizon is six months (to roughly April 2025). Of the three available expirations, December 2024 (~2.5 months out) is too short, and June 2025 (~9 months out) is too long; **March 2025** (~5–5.5 months out) is the closest match to the stated six-month concern and is therefore the appropriate choice.

Choice of strike: within the March 2025 expiration, the **4300 strike** is at-the-money (equal to the current index level), giving the most direct hedge against a "significant drop" from current levels, without paying for protection against a decline that has already partly happened (as a lower strike would) or leaving a gap near the current price unprotected (as a higher strike premium would otherwise imply).

Question 3 (continued)

(ii) Payoff if the S&P 500 falls 15%

Step	Calculation	Result
Portfolio value to hedge	100% index-tracking, full \$5,000,000	\$5,000,000
Contracts needed	$\$5,000,000 \div (4300 \times \$500)$	2.33 → round to 2 contracts
Index level after 15% fall	4300×0.85	3,655
Put intrinsic value (points)	$\text{Max}(4300 - 3,655, 0)$	645 points
Payoff per contract	$645 \times \$500$	\$322,500
Total payoff (2 contracts)	$2 \times \\$322,500$	\$645,000

Reconciliation — and a rounding direction worth noting. The portfolio loses $15\% \times \$5,000,000 = \$750,000$ in value. Using the exact (unrounded) hedge ratio of 2.3256 contracts, the put payoff is exactly \$750,000 — a perfect offset, as expected for an at-the-money put sized precisely to the exposure. Rounding the number of contracts **down** to the nearest whole number (2, since 2.3256 is closer to 2 than to 3) **under-hedges** the position: the payoff of \$645,000 falls short of the \$750,000 loss by \$105,000. This is the opposite direction of rounding to a similar hedging calculation in other papers in this series (where rounding up over-hedged); a candidate who instead rounded up to 3 contracts would get a payoff of \$967,500, which over-hedges by \$217,500. Both 2 and 3 contracts are defensible own-figure answers provided the rounding direction and its consequence (under- vs over-hedging) is clearly explained — this scheme uses nearest-whole-number rounding (2 contracts) as the primary answer, since that is the standard convention for "number of contracts needed" unless a bias toward full or excess protection is explicitly requested.

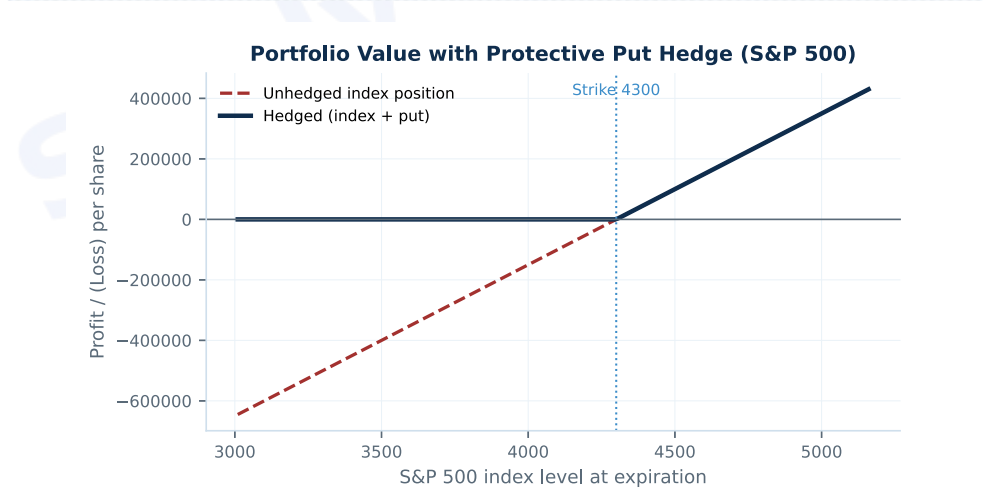


Fig. 1 — Value of the hedged vs unhedged S&P 500-tracking position around the March 4300 put

Mark Allocation — Part (a), 20 Marks

Tier	Marks	What separates this tier
16-20	16-20	Protective put correctly identified with the correct expiration (March, reasoned from the six-month horizon) and strike (4300, at-the-money) correctly justified; payoff calculation correct including contract sizing and a clear explanation of the rounding direction.
11-15	11-15	Correct strategy and broadly correct payoff, but expiration/strike choice not clearly reasoned, or the rounding direction/consequence not discussed.
5-10	5-10	Correct general strategy (buying puts) but wrong expiration/strike, or contract sizing omitted/incorrect.
0-4	0-4	Wrong strategy (e.g. selling options) or no credible attempt.

Question 3

50 Marks — Choice question: alternative to Question 2

(b) Required

10 Marks

Outline the five factors affecting option prices as specified in the Black-Scholes option pricing formula. Briefly discuss the effect of a change in each of these factors on the prices of both put and call options.

FULL MODEL ANSWER

Variable	Effect on Call price	Effect on Put price
Current underlying price (S)	Increases as S rises	Decreases as S rises
Exercise/strike price (K)	Decreases as K rises	Increases as K rises
Time to expiration (T)	Increases with more time	Generally increases with more time
Volatility of the underlying (σ)	Increases with volatility	Increases with volatility
Risk-free interest rate (r)	Increases as r rises	Decreases as r rises

Underlying price and strike price: A call gains value the further the underlying price is above the strike; a put gains value the further the underlying price is below the strike — the direct in-the-money effect for each option type, illustrated by the S&P 500 put above.

Time to expiration: More time gives more opportunity for a favourable price move before expiry, so both calls and puts are generally worth more with more time remaining, all else equal.

Volatility: Because option payoffs are asymmetric — losses capped at the premium paid, upside uncapped for the holder — higher volatility raises the value of both calls and puts, since it raises the probability of a large favourable move without symmetrically raising the cost of an unfavourable one.

Risk-free interest rate: A higher rate raises the cost of carrying the underlying and lowers the present value of the exercise price paid in future, increasing call values and decreasing put values.

Candidates who correctly identify all five variables and their qualitative direction of effect on both call and put prices should receive full credit — the numerical Greeks are not required at this level.

Mark Allocation — Part (b), 10 Marks

Tier	Marks	What separates this tier
8-10	8-10	All five variables correctly identified with accurate direction of effect on BOTH calls and puts.
5-7	5-7	Four or five variables identified with mostly correct direction, minor errors on one variable.
2-4	2-4	Two to three variables correctly identified, or direction of effect frequently confused between calls and puts.
0-1	0-1	Fewer than two variables correctly identified.

Question 3

50 Marks — Choice question: alternative to Question 2

(c) Required

20 Marks

"The creation of options and other derivative products which can be used to hedge uncertainty has removed a significant level of risk from the financial system". Critically discuss the above statement paying particular attention to any real world examples you may have encountered.

FULL MODEL ANSWER

This statement is only partly true, and a critical discussion needs to separate risk *transfer* at the individual level from risk in the financial *system* as a whole.

The case supporting the statement: For an individual hedger — such as the S&P 500 investor in part (a) above — buying a put option genuinely does reduce their exposure to a specific risk (a market decline), transferring that risk to the option writer in exchange for a premium. In this narrow sense, derivatives are a valuable risk-management tool that allows investors, companies, and financial institutions to manage exposures (currency risk, interest rate risk, commodity price risk) that they would otherwise have to bear in full.

The core critique — derivatives transfer risk, they do not eliminate it: A hedge does not make risk disappear from the financial system; it moves the risk from the hedger to the counterparty (the option writer, swap counterparty, or clearing house standing behind the trade). If that counterparty is itself highly leveraged, poorly capitalised, or has taken on large, concentrated derivative exposures, the risk has simply been relocated to a point in the system that may be less visible and less well capitalised to absorb it — this is a redistribution of risk, not a genuine reduction.

Real-world example — the 2008 Global Financial Crisis: Credit default swaps (CDS), a derivative designed to let investors hedge against bond or loan default, were written in enormous volume by AIG's Financial Products division against mortgage-backed securities. Rather than diversifying risk safely across the system, this concentrated enormous counterparty risk in a single institution; when the underlying mortgage securities began to fail, AIG could not meet its obligations, and the U.S. government was forced into an \$180+ billion bailout to prevent AIG's failure from cascading through the many banks that had bought CDS protection from it — a clear example of derivatives amplifying, rather than removing, systemic risk.

Real-world example — Long-Term Capital Management (1998): LTCM used highly leveraged derivative positions (including swaps and options) intended to exploit small pricing anomalies with hedged, low-risk strategies; when markets moved against several of its positions simultaneously during the 1998 Russian financial crisis, the fund's scale and interconnectedness with major banks meant its near-collapse threatened broader financial stability, requiring a Federal Reserve-organised bailout by a consortium of banks.

Real-world example — Barings Bank (1995): A single trader (Nick Leeson) built up large, unauthorised, unhedged derivative (futures and options) positions on the Nikkei 225 index; when the market moved sharply against these positions following the Kobe earthquake, the resulting losses (around £827 million) exceeded the bank's entire capital base, collapsing a 233-year-old institution — illustrating how derivatives used for speculation rather than genuine hedging can destroy, rather than protect, an institution.

Balanced conclusion

Distinguish hedging from speculation, and individual from systemic risk: Used properly for genuine hedging by well-capitalised, transparent counterparties, derivatives do provide real risk-management value at the individual level. However, the historical record — AIG/CDS in 2008, LTCM in 1998, Barings in 1995 — shows that derivatives have repeatedly been a source of amplified, concentrated, and sometimes systemic risk when used for leveraged speculation, when counterparty risk is underappreciated, or when positions are large relative to the capital base backing them.

Overall assessment of the statement: The claim that derivatives have removed 'a significant level of risk from the financial system' is an overstatement; a more accurate characterisation is that derivatives have changed the *distribution and visibility* of risk within the system — reducing risk for individual hedgers who use them prudently, while at various points in financial history concentrating and amplifying risk at the systemic level, particularly where regulation, transparency, and capital adequacy have not kept pace with the scale and complexity of derivative markets (a key motivation behind post-2008 reforms such as mandatory central clearing for many derivative contracts).

Candidates may cite different real-world examples (e.g. Orange County 1994, JPMorgan's 'London Whale' 2012). What matters is a genuinely critical discussion — not simply agreeing or disagreeing with the statement — that distinguishes individual-level hedging benefits from systemic-level risk concentration, supported by at least one well-explained real-world example.

Mark Allocation — Part (c), 20 Marks

Tier	Marks	What separates this tier
16–20	16–20	Genuinely critical discussion distinguishing individual hedging benefits from systemic risk concentration, with at least two well-explained real-world examples (e.g. 2008 GFC/AIG, LTCM, Barings) and a clear, reasoned overall assessment of the statement.
11–15	11–15	Good critical discussion with at least one real-world example, but the individual-vs-systemic distinction underdeveloped.
6–10	6–10	Statement discussed in general terms (derivatives are useful for hedging) with little critical challenge or no real-world examples.
0–5	0–5	Statement simply accepted at face value with no critical engagement, or no meaningful discussion.

Question 3 Total: 50 Marks